ENTERPRISE DATABASE DESIGN, IMPLEMENTATION, AND WEB INTEGRATION

Course: Database Systems

Project Title: Wave Rentals Administrative Database System

Institution: Ho Technical University

Group Number: Group Six (6)

Group Members:

- **Zartey Walter** (0324080051)
- **Kordorwu Emmanuel** (0324080011)
- **Daniel Lawoekpor** (0324080562)
- **Agbo Divine** (0324080037)
- **Dzimale Micheal Komla** (0324080526)
- **Tsri Sylvia** (0324080251)
- **Arthur Elizabeth Pokua** (0324080433)
- **Kushitor Horlanyo Benjamin** (0324080492)
- **Ohene Timothy Atu** (0324080506)
- **Ofori Daniel Oheneasa** (0324080367)

Part 1: Business Requirements Specification

1.1 Company Profile

Wave Rentals is a specialized car rental company based in Ho, Volta Region, managing a diverse fleet of passenger vehicles. The enterprise serves tourists, business executives, and local residents. To optimize transaction speeds, secure business records, and improve administrative efficiency, Wave Rentals is moving away from manual ledger tracking to a secure, automated database system.

1.2 Core Operations

The database system tracks and manages five critical operational branches of the business:

1. **Customer Profiles:** Logging names, unique driver's licenses, and contact information.
2. **Fleet Management:** Keeping records of each car's unique identifier, brand, model, license plate, and availability status.
3. **Staff Tracking:** Documenting internal employees authorized to process rental contracts.
4. **Rental Agreements:** Logging transaction checkouts, assigning customers to vehicles, designating the authorizing employee, and capturing checkout/check-in dates.
5. **Financial Transactions:** Monitoring payments linked directly to rental bookings, tracking amounts, payment dates, and payment methods.

1.3 System Business Rules

To enforce operational integrity, the application strictly adheres to the following logical rules:

- A **Customer** must provide a unique driver's license number and contact phone number to register.
- A **Car** must have a unique physical license plate registered in the system. Newly registered cars default to an 'Available' status.
- An **Employee** must be registered within the staff table before they can officially process any rental agreements.
- A **Rental** transaction can only be authorized for an existing customer, car, and employee. The checkout date cannot be null.
- A **Payment** cannot exist without being mapped directly to an active rental agreement. It must capture a decimal monetary value and the transaction channel (e.g., Mobile Money, Cash, Card).
-

Part 2 & 3: Entity Types, Attributes, and Data Integrity Constraints

Below is the structured data dictionary establishing technical parameters, key constraints, nullability, and domain constraints for the Wave Rentals database:

1. Customer Table (customer)

Enforces distinct identities for clients checking out vehicles.

- customerID (**INT**, **PK**, AUTO_INCREMENT): Unique, system-generated primary identifier.
- Name (**VARCHAR(100)**, NOT NULL): The full name of the customer.
- License_No (**VARCHAR(50)**, NOT NULL, UNIQUE): Driver's license number used for verification.
- phone_No (**VARCHAR(15)**, NOT NULL, UNIQUE): Contact phone number.

2. Car Table (Car)

Maintains active inventory status of the vehicle fleet.

- VehicleID (**INT**, **PK**, AUTO_INCREMENT): Unique inventory ID.
- License_plate (**VARCHAR(50)**, NOT NULL, UNIQUE): Official physical vehicle plate number.
- Model (**VARCHAR(50)**, NOT NULL): The specific car model.
- Brand (**VARCHAR(50)**, NOT NULL): Car brand/manufacturer.
- Status (**VARCHAR(20)**, DEFAULT 'Available'): Tracks if the car is currently 'Available' or 'Rented'.

3. Employee Table (Employee)

Identifies operational staff members processing bookings.

- EmployeeID (**INT**, **PK**, AUTO_INCREMENT): Unique staff identifier.
- Fname (**VARCHAR(50)**, NOT NULL): First name.
- Lname (**VARCHAR(50)**, NOT NULL): Last name.
- position (**VARCHAR(50)**, NULL): Employment role within Wave Rentals.
- sex (**VARCHAR(10)**, NOT NULL): Employee gender.

4. Rental Table (Rental)

Handles the central business transaction mapping customers, cars, and authorizing staff.

- RentalID (**INT**, **PK**, AUTO_INCREMENT): Unique rental agreement number.

- CustomerID (**INT**, **FK**, NOT NULL): References customer.customerID.
- VehicleID (**INT**, **FK**, NOT NULL): References Car.VehicleID.
- EmployeeID (**INT**, **FK**, NOT NULL): References Employee.EmployeeID.
- RentalDate (**DATE**, NOT NULL): The rental contract commencement date.
- ReturnDate (**DATE**, NULL): The physical return check-in date.

5. Payment Table (Payment)

Manages billing transactions tied directly to rental records.

- PaymentID (**INT**, **PK**, AUTO_INCREMENT): Unique payment reference number.
- RentalID (**INT**, **FK**, NOT NULL): References Rental.RentalID.
- paymentDate (**DATE**, NOT NULL): Date payment was completed.
- Amount (**DECIMAL(10,2)**, NOT NULL): Transaction payment amount.
- Pamethod (**VARCHAR(50)**, NOT NULL): Channel (e.g., Mobile Money, Cash).

Part 4, 5, & 6: Diagrams

- **For Part 4 (Isolated Schema):** From the slides
- **For Part 5 (ER Diagram):** From the slides
- **For Part 6 (Logical Schema):** From the slides

Part 7: Physical Database Implementation & Population

Below are the complete SQL scripts written and executed using the MySQL engine inside XAMPP phpMyAdmin to build and seed the database.

7.1 Data Definition Language (DDL) Script

SQL

```
CREATE DATABASE IF NOT EXISTS wave_rentals;

USE wave_rentals;
```

```sql
-- Create Customer Table
CREATE TABLE customer (
    customerID INT AUTO_INCREMENT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    License_No VARCHAR(50) NOT NULL UNIQUE,
    phone_No VARCHAR(15) NOT NULL UNIQUE
);

-- Create Car Table
CREATE TABLE Car (
    VehicleID INT AUTO_INCREMENT PRIMARY KEY,
    License_plate VARCHAR(50) NOT NULL UNIQUE,
    Model VARCHAR(50) NOT NULL,
    Brand VARCHAR(50) NOT NULL,
    Status VARCHAR(20) NOT NULL DEFAULT 'Available'
);

-- Create Employee Table
CREATE TABLE Employee (
    EmployeeID INT AUTO_INCREMENT PRIMARY KEY,
    Fname VARCHAR(50) NOT NULL,
    Lname VARCHAR(50) NOT NULL,
    position VARCHAR(50),
    sex VARCHAR(10) NOT NULL
);
```

```sql
-- Create Rental Table (With Foreign Keys & Cascading)
CREATE TABLE Rental (
   RentalID INT AUTO_INCREMENT PRIMARY KEY,
   CustomerID INT NOT NULL,
   VehicleID INT NOT NULL,
   EmployeeID INT NOT NULL,
   RentalDate DATE NOT NULL,
   ReturnDate DATE NULL,
   FOREIGN KEY (CustomerID) REFERENCES customer(customerID) ON DELETE CASCADE,
   FOREIGN KEY (VehicleID) REFERENCES Car(VehicleID) ON DELETE CASCADE,
   FOREIGN KEY (EmployeeID) REFERENCES Employee(EmployeeID) ON DELETE CASCADE
);

-- Create Payment Table
CREATE TABLE Payment (
   PaymentID INT AUTO_INCREMENT PRIMARY KEY,
   RentalID INT NOT NULL,
   paymentDate DATE NOT NULL,
   Amount DECIMAL(10,2) NOT NULL,
   Pamethod VARCHAR(50) NOT NULL,
   FOREIGN KEY (RentalID) REFERENCES Rental(RentalID) ON DELETE CASCADE
);
```

7.2 Data Manipulation Language (DML) Script

SQL

```sql
USE wave_rentals;

-- Populate Customers
INSERT INTO customer (Name, License_No, phone_No) VALUES
('Kofi Mensah', 'GHA-829102-A', '0244123456'),
('Ama Serwaa', 'GHA-930219-B', '0555987654'),
('Selorm Alornyo', 'GHA-481029-C', '0201122334');

-- Populate Fleet Cars
INSERT INTO Car (License_plate, Model, Brand, Status) VALUES
('VR-101-26', 'Corolla', 'Toyota', 'Rented'),
('VR-302-26', 'Civic', 'Honda', 'Available'),
('VR-505-26', 'Elantra', 'Hyundai', 'Available');

-- Populate Staff Members
INSERT INTO Employee (Fname, Lname, position, sex) VALUES
('Mawuli', 'Koku', 'Rental Agent', 'Male'),
('Esenam', 'Abla', 'Manager', 'Female');

-- Populate Rental Agreements
INSERT INTO Rental (CustomerID, VehicleID, EmployeeID, RentalDate, ReturnDate) VALUES
(1, 1, 1, '2026-07-01', '2026-07-05'),
(2, 3, 1, '2026-07-10', NULL);
```

```sql
-- Populate Payments

INSERT INTO Payment (RentalID, paymentDate, Amount, Pamethod) VALUES

(1, '2026-07-01', 1200.00, 'Mobile Money'),

(2, '2026-07-10', 850.00, 'Cash');
```

Part 8: Web Interface Integration

To make this enterprise operational layer user-friendly for Wave Rentals administrative staff, a PHP-powered web portal was deployed. It serves as a real-time system with direct database driver integration.

8.1 Technical Architecture

- **Frontend (HTML5 / CSS3):** Designed as a responsive grid displaying a data entry form on the left side and an operations directory table on the right side.
- **Database Connection (PDO):** Built with security in mind using **PHP Data Objects (PDO)**, utilizing prepared statements to shield the application from SQL injection vulnerabilities.
- **System Action Features:**
    1. **Data Entry Action (Customer Registration Form):** Feeds new customer registrations securely into the customer database table. The controller features real-time validation, catching MySQL unique key constraint conflicts (such as duplicated driver's license entries or phone numbers) and rendering them as polite, user-friendly, and dismissible alerts.
    2. **Data Retrieval Action (Dynamic Filter Directory):** Pulls live transactional data by executing cross-table relational joins. It features an **active search bar** allowing administrators to dynamically filter rental transactions by client name, vehicle model, or license plate.